

1 Invention and development

THOMAS LILEY

Introduction

The word ‘saxophone’ means ‘the sound of Sax’ – specifically that of Adolphe Sax. The Greek word ‘phone’, according to the *Oxford English Dictionary*, relates in particular to vocal sounds, so we should not be surprised that the saxophone is often described as a ‘singing’ instrument. In fact, the saxophone is the most flexible and expressive of musical instruments, exceeded, perhaps, only by the human voice. The human voice, of course, is capable of sounds as varied as cheerleading and Schubert lieder; it is capable of producing guttural sounds and fine-spun eloquence, of rabble-rousing and of inspiring. The saxophone is similar in its potential to move people, both viscerally and emotionally.

As a ‘singing’ instrument, the saxophone is unmatched by its mechanical counterparts. This is often reflected in its classical repertoire, but there can be little doubt that the saxophone is the pre-eminent jazz instrument. If jazz first came from the work-shouts and the blues sung in the nineteenth century, then it is natural that a ‘singing’ instrument such as the saxophone should be well suited to jazz and popular music. What is remarkable is that the instrument took so long to be adopted by jazz musicians.

The saxophone is the invention of a single man, Adolphe Sax. Both a thinker and a doer, he had the genius to conceive a new and versatile instrument, the practical background to bring his theories to fruition and the foresight to create the mechanisms necessary to ensure that it would become an important part of the musical world.

Adolphe Sax

Born in Dinant, Belgium, on 6 November 1814, Adolphe Sax was the first child of Charles and Maria Sax (christened Antoine Joseph, he was called Adolphe from childhood). Shortly afterwards the family moved to Brussels

and Charles began a new and successful career which was to profoundly influence his son – he turned from the manufacture of cabinets and tables to the manufacture of clarinets and brass instruments.



Fig. 1.1 Dinant, Belgium, birthplace of Adolphe Sax

Charles was appointed Instrument Maker to the Court of the Netherlands, and is perhaps best remembered for his version of the *cor omnitonique*. The instrument was intended as a solution to the numerous and cumbersome crooks of the hand horn, but lost out to the lighter-weight and more efficient valved horn.

Adolphe was often to be found in his father's workshop and at an early age learned the skills necessary to the profession. In addition to the instruction he received from his father, he also received a formal education

and music instruction at the Brussels Conservatory. He studied voice and flute; later he also studied clarinet. It is said that Sax could have pursued a successful career as a clarinettist, but his interest in overcoming the instrument's acoustical shortcomings kept returning him to the workshop to seek solutions.

Sax's skill as an instrument maker grew rapidly, and his work soon began to attract favourable comment. Horwood suggests that Charles was principally concerned with the production of instruments for sale and that he left various experimental projects to Adolphe. These experiments included the development of valved bugle-horns – the saxhorns – and the combination of a conical metal tube with a single-reed mouthpiece – the saxophone.¹

Another important success of this period was the construction of a vastly improved bass clarinet. It was the bass clarinet which first brought Adolphe Sax to Paris in 1839, where he sought out Isaac Dacosta, solo clarinet at the Paris Academy of Music and creator of his own improved bass clarinet. Sax persuaded Dacosta to listen to him perform the solo from Meyerbeer's *Les Huguenots* and to acknowledge the superiority of Sax's instrument. During the course of this first visit Sax met many prominent musicians, including Berlioz, Halévy, Kastner and Meyerbeer. These influential contacts were to prove crucial to Sax when, in 1842, he returned to Paris for good.

Sax's decision to leave Brussels was sealed by officials of the Brussels Exhibition of 1841. Among the items which Sax prepared for the Exhibition were several clarinets and, Sax's friend Georges Kastner tells us, a saxophone. This would have been the first public presentation of the saxophone except that, Kastner continues, the instrument 'was sent flying with a kick by an unknown person at a time when the inventor, Adolphe Sax, was away'.² The examining committee recommended the young craftsman's display for the Premier Gold Medal, the Exhibition's highest award. The recommendation, however, was rejected by the Central Jury, which declared against his youth: if Sax were to receive the first prize this year, there would be nothing to give him the next year. Sax's response was typical of the man: 'If I am too young for the gold medal, I am too old for the silver'. Paris beckoned.³

Shortly after this incident Sax was visited by an influential aide to the French king Louis-Philippe. Lieutenant General Comte de Rumigny spoke to Sax of ambitious plans to revitalise French military bands, and saw the Belgian's work as a means to that end. Sax was further encouraged by a

letter from composer Jacques Halévy, whom he had met in Paris. The self-confident instrument maker left Brussels behind and moved to Paris with his ideas, his creations and his hopes in the spring of 1842.

Successes and opposition

Sax's arrival was heralded by Hector Berlioz in the 12 June 1842 issue of the highly regarded *Journal des débats*. In a column entitled 'Adolphe Sax's Musical Instruments' Berlioz introduced the young inventor to the French public. After a few paragraphs in which Berlioz describes the development of musical instruments, he presents:

M. Adolphe Sax of Brussels, whose work we have just examined. . . .He is a man of penetrating mind; lucid, tenacious, with a perseverance against all trials, and great skill. . . .He is at the same time a calculator, acoustician, and as necessary also a smelter, turner and engraver. He can think and act. He invents and accomplishes.⁴



Fig. 1.2 Bust of Adolphe Sax

There follows a description of several of Sax's instruments, including the first detailed account of the saxophone. Berlioz closes with the prediction that 'Composers will be much indebted to M. Sax when his instruments

come into general use. May he persevere; he will not lack support from friends of art.’⁵

One of the results of this announcement was a public concert that same month at the Paris Conservatoire, at which Sax played proficiently and talked eloquently about his instruments. Sax met and was befriended by many of the most distinguished musicians in Paris, including Auber, Habeneck, Spontini and Donizetti. Funds were gathered for the establishment of the Adolphe Sax Musical Instrument Factory at No. 10 Rue Saint Georges.

He also aroused the beginnings of a bitter enmity between himself and the established instrument makers of Paris. They had been put on notice that the young Belgian would undoubtedly challenge their businesses, and his close association with so many prominent musicians clearly threatened their success. Thus began a round of threats, thefts and legal battles (and, possibly, even an attempt on his life) that would pursue him for decades. Berlioz, in a letter dated 8 October 1843, wrote:

It is scarcely to be believed that this gifted young artist should be finding it difficult to maintain his position and make a career in Paris. The persecutions he suffers are worthy of the Middle Ages and recall the antics of the enemies of Benvenuto, the Florentine sculptor. They lure away his workmen, steal his designs, accuse him of insanity, and bring legal proceedings against him. Such is the hatred inventors inspire in rivals who are incapable of inventing anything themselves.⁶

Among the tactics employed by these adversaries was an engineered opposition to the general acceptance of Sax’s instruments in orchestras. Many important performers served as consultants to instrument manufacturers and refused to play on an instrument by a rival maker. One such incident occurred in 1843, when, while writing his opera *Dom Sébastien*, Gaetano Donizetti prepared a part for the Sax bass clarinet. The principal clarinettist of the Opera orchestra threatened that, if the instrument were to be used, the orchestra would walk out. Donizetti, despite his high regard for Sax, was forced to bow to the musicians’ demands.

The animosity between Sax and his rivals intensified when the French government made an announcement that would affect every wind instrument maker in France: the reform of the French military bands. The abysmal state of French military music was an embarrassment to French patriotism and prestige. To address the situation a commission consisting of musical, military and acoustical representatives, and headed by General de Rumigny, met on 25 February 1845. Manufacturers were requested to submit

instruments for consideration by the commission; only Sax gave a full response to the invitation.

Perhaps the most auspicious event of the commission's activities occurred on 22 April of that year when two bands performed in competition before an audience, estimated at 20,000, gathered on the Champ du Mars in Paris. One band, under Michele Carafa, director of the *Gymnase de musique militaire* and a member of the commission, represented a conservative reorganisation of the existing ensembles. The other band, directed by M. Fessy, demonstrated fully Sax's comprehensive reforms and his instruments – the saxophone and, especially, the saxhorn.

Each band was to consist of forty-five players and would perform arrangements of music by Adolphe Adam and a work selected by their conductor. The audience declared Sax's ensemble the clear winner, and the commission took the results under advisement as part of the process. The final report, submitted to the Minister of War on 9 August 1845, was a distinct victory for the Belgian – the government had declared a near monopoly mandating the use of his instruments.

Rival instrument makers had been placed in an untenable position. Faced with financial destruction, they consolidated against the common enemy from Dinant. *L'Association générale des ouvriers en instruments de musique* (the United Association of Instrument Makers) was established to protect their interests.

One of the Association's first lawsuits opposed Sax's application to patent the saxophone. Various contradictory lines of attack were made: the instrument does not exist; if it does exist, it is a completely unmusical invention; and, in any event, it is not original because it already exists in other forms. Among the presumed antecedents were the 'English tenoroon', a bass clarinet by Desfontenelles, and the German bathyphone by Wieprecht. In another tactic several saxophones were purchased and sent to other countries; Sax's identification was removed and the instruments were then re-engraved to indicate foreign manufacture. These forgeries were poorly executed and quickly revealed as a ruse.

Of greater legal consequence was the contention that, because the saxophone had been performed before a large public audience during the contest on the Champ du Mars, it was invalid for patent. Here the Association could cite a legal precedent. Sax's response was direct and dramatic: he challenged his opponents to create the instrument (which did or

did not exist) known as the saxophone and he withdrew his patent request for a year to give them ample time to do so. The plaintiffs were unable to create a rival instrument.

To respond in the form of a challenge was typical of Sax. Also typical was the impatience that led him to reapply for the patent shortly before the year's respite had elapsed. Sax was granted a patent for the saxophone family on 22 June 1846, sometimes referred to as the birth date of the saxophone. The Association, however, was not through.

The Revolution of 1848 provided the Association with a new opportunity when King Louis-Philippe was forced to abdicate. Louis-Philippe's staff, which included many of Sax's most influential supporters, fled with him. Sax's lengthy relationship with the king was now a serious liability, and the 1845 reforms of military bands were revoked by the new government in a bill introduced by the Association. Orders for new Sax instruments were cancelled and deliveries were returned. The closure of the factory seemed inevitable until an anonymous benefactor stepped forward; his generosity unintentionally led the impulsive Sax to bankruptcy.

The anonymous business associate, concerned by rising unemployment in Paris during the turbulent period, gave Sax 30,000 francs to pay his workmen. Because no receipt was requested, Sax understood the sum to be a gift rather than a loan. The benefactor died in 1852, with the money unaccounted for in his estate. Sax rashly acknowledged in writing that he had received the money as a loan and was immediately ordered to repay the debt within twenty-four hours. He fled to London and pleaded his case by mail. His pleas were ignored and he returned to Paris and financial ruin.

Battles and bankruptcy

The series of lawsuits and appeals begun in 1846 would consume Sax's physical and financial resources throughout the remainder of his life. Although there seems to be no evidence that the Association was involved in the circumstances which led to Sax's bankruptcy, they were undoubtedly pleased by the turn of events. Sax's ability as an inventor was not matched by his acumen as a businessman nor as a litigant.

The Revolution of 1848 greatly affected Sax's legal status. Whereas in late 1847 his patents had been upheld by the courts, by 1849 an appeal by the Association rendered the patents for *bugles-à-cylindres* and saxotrombas

void. The saxophone patent, however, was sustained. Appeals and counter-appeals by both Sax and the Association led to an eventual conclusion at the Imperial Court at Rouen in 1854. Eight years after the original lawsuit was filed the high court ruled completely in favour of Sax, declaring that his patents were valid and that he was entitled to damages and reimbursements from the Association. None the less, damage had been done. Time and money which could have been spent creatively and productively had been expended on seemingly endless legal procedures.

Undoubtedly Sax had several reasons to be optimistic at the apparent conclusion of his legal battles. In April 1854 he was appointed Musical Instrument Maker to the Household Troops of Emperor Napoleon III. The emperor further showed his favour by assisting Sax in his bankruptcy. Arrangements were made with Sax's numerous creditors, allowing him to reopen his factory on the Rue Saint Georges, now at No. 50.

Despite his vindication in the courts and the renewed support from the emperor, Sax was adversely affected by his years of legal battles. Perhaps his natural contentiousness was intensified by the slanderous misrepresentations to which he was subjected, or perhaps he felt that he now understood the legal system and could use it to his advantage. Whatever the cause, there can be little doubt that his health was also an important factor.

Since 1853 Sax had been concerned about a dark spot on his lip which had blackened and grown in size. The tumour was diagnosed as malignant; surgical removal of the tumour, including lip and part of the jaw, seemed inevitable. The cancer obstructed Sax's throat to the extent that it became necessary for him to be fed by means of a tube. The recommended operation was delayed by the lawsuits, and Sax cast about for a cure that would avoid disfiguring surgery. He was introduced to Dr Vries, an Indian doctor in Paris, who had achieved some success in treating incurable ailments, and in 1859 the black spot began to decrease until it had completely and miraculously disappeared.

It was during this troubled period that Sax made the decision to return to the courts, but now as a plaintiff. In 1859 Sax sued Pierre Louis Gautrot for a perceived infringement of his saxophone patent. In 1856 Gautrot had patented a family of instruments by M. Sarrus. The instruments, called sarrusophones, were too similar to the saxophone in Sax's judgement. Like the saxophone, the sarrusophone is conical in shape, has the same written range from b to f3,⁷ and is grouped in similar keys and registers. The sounds

produced by the two instrument families, however, are markedly different, and the court ruled against Sax, deeming the instruments to be completely different.

Sax again returned to the courtroom as a plaintiff in 1866, and this time was successful in a suit which did him no good. A well-known opera singer, Marie-Constance Sass, had grown dissatisfied with her name and had changed it to 'Sax'. The inventor's objections, upheld by the court, led her to suggest 'Saxe' as an alternative. This was also unacceptable to Sax and to the court, whose ruling gave Sax a victory of doubtful significance.

Other events were of more value to Sax. In 1847 he was appointed Musical Director of the stage brass band at the Opera, a position he would hold until his death, and which would become his sole reliable source of income.⁸ On 7 June 1857, Sax was asked to institute a saxophone class at the Paris Conservatoire; he remained in that position until the catastrophic war of 1870 between France and Prussia. He received medals and awards from the emperor and around this time became the father of two sons and a daughter. Charles, born in October 1856, died in infancy in 1858. Adele, born 29 November 1858, lived until 1938, and his son Adolphe Edouard, born 29 September 1859, lived until 1945. Adolphe Edouard continued the family business until it was sold to the Henri Selmer Company in 1928.⁹

Sax continued to present his instruments at various exhibitions and in 1867, at the Paris International Exhibition, was awarded the Grand Prize. Horwood reports that the display included an example of every musical instrument Sax had invented or improved. The centrepiece was an exquisite gold-plated alto saxophone.¹⁰

This distinction may have been the apex of Sax's life. Because Sax had regained solvency since his bankruptcy of 1852, he was astonished to learn that because the recovery had not been officially recorded by the Commerce Court, he was dismissed from the Legion of Honour. The decoration, bestowed by the emperor in 1849, was one of Sax's most distinguished awards and he wore the cross with justifiable pride. Its loss in 1860 because of an administrative omission humiliated him.

The emperor himself was deposed after the French surrender at Sedan to the Prussians in September of 1870, costing Sax his most influential ally. Many of Sax's close colleagues passed away in the years between 1867 and 1871 – Kastner in 1867, Rossini in 1868, Berlioz in 1869, and Auber and

Fétis in 1871. Their counsel and support would be sorely missed in the years ahead.

By this time he was estranged from his brother Alphonse, also an inventor and instrument maker, with whom he had quarrelled at the 1862 Exhibition in London. His father, Charles, had died in 1865, and in 1871 his brother Charles-Joseph passed away; both had provided important assistance at the factory on the Rue Saint Georges.

Sax had been unable to take full benefit of the opportunities his exclusive patents had afforded him. When his patents ceased in the mid-1860s, other instrument manufacturers moved quickly on the expired patents, and orders to the Sax factory diminished. As his financial situation worsened, he was struck another blow.

The French defeat at Sedan caused a number of financial rescissions, including the close of the saxophone class at the Conservatoire in 1870; it would not be reinstated until 1942. Not only was Sax deprived of an important source of income, the future of the saxophone was endangered. Realising that without teachers there could not be students for the new instrument, Sax offered to continue to teach without fee; his proposal was rejected. In 1873 Sax, for the second time, was declared bankrupt.

The Rue Saint Georges factory was shut down, its contents sold to appease creditors. Sax's acclaimed collection of 467 musical instruments was sold in December 1877; many examples are still in museums in Brussels and Paris.¹¹

All that remained were the lawsuits and the position of Musical Director at the Opera. The Association had continued to contest the judgement of 1854; throughout the 1880s Sax persevered in his battle for the compensation due him but was never to receive the reparations he had been awarded.

A poignant final petition came from Sax in 1887 in the journal *La Musique des Familles*. In his 'Appeal to the Public' the seventy-two-year-old Sax asks once again for justice in his attempts to gain restitution from the Association after twenty-six years of waiting. This remarkable document brought no response from the legal system.¹² It did succeed, however, in motivating several musicians, led by composer Paul Lacôme, in their own appeal on Sax's behalf. The request resulted in a modest pension for the aged inventor.



Fig. 1.3 Alto saxophone by Adolphe Sax, c. 1857

In keeping with the turbulence of his life, there is also controversy about the date of Sax's death. Several sources, including *Grove*, Horwood, and Ronkin, cite 4 February 1894 as the date of his death. Gee gives, simply,

1894, while Kochnitzky says that ‘Sax lived to the age of eighty’.¹³ Because Sax was born 6 November 1814, this implies that he lived past 6 November in 1894. The correct date, found in *Baker’s*, Deans, and Haine, is 7 February 1894.

The saxophone: theories of invention

Various theories surround the creation of the saxophone, many of which are as complex and confusing as the lawsuits which plagued its creator. Several of these theories seem to have derived some credibility from their mention in the lawsuit of 1846 in which Sax’s patent of the saxophone was challenged and, ultimately, upheld. There are, in fact, precedents for Sax’s creation, of which his challengers seem to have been unaware.

Frederick Hemke mentions an Argentine instrument cited by Bessaraboff. The instrument is made of a cow’s horn whose tip is shaped to resemble a single-reed mouthpiece with a thin reed of bone bound by a silk thread. It is said to have been derived from a similar and even older instrument.¹⁴

Another presumed ancestor of the saxophone is the alto fagotto. This instrument, invented by William Meikle of Strathaven, in Lanarkshire, and made by George Wood of London in 1830, is ‘a conical wooden tube shaped like a bassoon, but sounded by means of a single-reed attached to the end of a bassoon-like crook’.¹⁵ Referred to by Sax’s rivals in their suits, the instrument has received several names, including caledonica, tenoroon and dolciano. There is a photograph in Bessaraboff.¹⁶

An instrument created in 1807 by Desfontenelles, a clock maker in Liseux, was also presented as a predecessor to the saxophone. The joining of a wooden conical tube with a clarinet mouthpiece was untested until Jaap Kool played the instrument in about 1930. He discovered that, unlike the saxophone which overblows an octave, the Desfontenelles instrument produces twelfths, as does the clarinet, and therefore is not a prototype of the saxophone. The instrument is in the collection at the Paris Conservatoire¹⁷ and is pictured in Horwood.¹⁸

A still-less likely challenger was the bathyphone. Its creator, Friedrich Wilhelm Wieprecht, is credited with the invention of the tuba in 1835 and an improved contrabassoon. The bathyphone is described by Carse as a contrabass clarinet with a ‘wide tube doubled like a bassoon, and the eighteen note-holes were all covered by keys working on rod-axles, and

arranged in two groups. The metal bell pointed upwards, and at the upper end the bore was continued in a metal tube or crook bent over towards the player.’¹⁹ The instrument, therefore, has no similarity to the saxophone, but the antagonism between the two inventors provided fuel for the legal fires. Baines provides a photograph.²⁰

One other candidate proposed as a predecessor to the saxophone is the Hungarian *tárogató*. This primitive instrument has a conical wooden bore and may be seen in Baines.²¹ It produced its sound with a double reed until it was modernised by W. J. Schunda of Budapest around 1900, long after the saxophone’s invention. In addition to providing the instrument with a single-reed mouthpiece, Schunda also gave it a complete key system.²²

Opinions of how Sax invented the saxophone fall into three categories according to Hemke: (1) Sax was searching for a clarinet that could play octaves rather than twelfths; (2) he substituted the cup mouthpiece of the ophicleide with a single-reed mouthpiece; and (3) he experimented with a single-reed mouthpiece on a rudimentary bassoon. Still others say that the invention happened by accident or by mistake.²³ Sax’s son, Adolphe Edouard, said in 1925 that his father’s work was intentional and well thought out, the ‘construction of a brass instrument equipped with a vibrating reed and adapted to a parabolic cone’.²⁴

Similar confusion exists concerning the exact date of the saxophone’s invention. The suggested year of invention varies from sometime around 1840 to 1846, the year of the saxophone patent. Sax himself did not indicate the date, but Maurice Hamel wrote that his father, Henry, a very close friend of the inventor, had said that Sax created the saxophone in 1838. Hamel states that the saxophone came into existence soon after Sax’s new bass clarinet, which was presented at the Belgian Exposition of 1835. Hamel’s handwritten recollections give the most credible date for the invention of the saxophone.²⁵

Early uses

While one may not be surprised that Sax and his supporters found many opportunities to present the new instrument in the fertile musical life of Paris, it is interesting to note how quickly the saxophone spread to other countries. Frederick Hemke and Harry Gee provide us with the names of several saxophone pioneers who travelled to a remarkable number of

countries and who laid the foundation for today's performers. Adolphe Sax, of course, was the first to play his creation, but the future of the instrument depended on its being heard by large audiences in a variety of venues. Only names remain for many of these saxophonists – Auroux, Beeckman, Cordier, Demange, Hernandez, Lecerf and Printz – but three individuals should continue to be remembered for their exploits and their artistry.

Louis-Adolphe Mayeur was born in Belgium in 1837, studied clarinet with Klosé at the Paris Conservatoire and saxophone with both Klosé and Sax; he died in 1894. He was regarded as the most brilliant saxophone performer of his era, appearing as soloist with the Brussels Opera and, from 1871, the Paris Opera. He wrote a large number of transcriptions, original works and study material, some of which remain in print.²⁶

Mayeur was one of several renowned musicians hired by Sax in 1864 to travel throughout France, Holland and Belgium demonstrating Sax's instruments in concert. We can judge his ability from a review by Johannes Weber in *Le Temps* (4 April 1867), in which Mayeur is described as an excellent artist, '... joining a remarkable beauty of sound to a style equally able to present large and expressive melodies as well as executing the most brilliant and most difficult passages'.²⁷

A more shadowy and altogether fascinating performer is the saxophonist known only as Souallé. Born in the north of France, his birth and death dates are unknown. He is credited with the first solo appearance of the saxophone in London in 1850, where he performed with the conductor and impresario Louis Antoine Jullien.²⁸ He is also undoubtedly the first saxophonist to appear in such distant lands as India, Australia, Java and China, performing as 'Ali-Ben-Sou-Alle' on a saxophone he called the 'turcophone'. He also toured throughout Europe. Souallé wrote several works for saxophone that recall his world travels (such as *Souvenirs de Java*, *Souvenirs de Shanghai*) and a few other works; none is currently in print.

There can also be little doubt that he was an artist as well as a world traveller. Souallé is the subject of a laudatory review by Berlioz in the 13 April 1851 issue of *Journal des débats*:

Mr Souallé, who recently returned from London, produced a great sensation by performing on Sax's masterpiece, the saxophone. *This was the first time the saxophone, with all its advantages, has been heard in Paris* [emphasis added]. This instrument possesses incomparable and expressive qualities; the trueness and beauty of sound which can be produced when one really masters the technique are such that it can, in slow pieces, challenge the finest singers.²⁹

Henri Wuille was also hired by Jullien, who took him to England in 1852, and with whom he performed until 1856. While with Jullien he toured the United States as first clarinettist and saxophone soloist. He is credited with the first saxophone solo performance in the United States in a concert presented in New York on 19 December 1853.³⁰ Born in Belgium in 1822, Wuille studied clarinet with Valentin Bender, who also taught Sax. After his tours with Jullien, Wuille eventually settled in Strasbourg, where he taught at the Conservatoire. He died in Baden, Germany, in 1871.

In the *Revue et Gazette musicale* (11 July 1858), Kastner made the following comments about Wuille's abilities after hearing him perform a *Fantasy on Themes from 'Martha'* with the orchestra in Baden in 1858:

Adding perfect taste to a marvellous ability and brilliant execution to the solid qualities of a good musician, Mr Wuille enjoys a justly acquired universal reputation. Probably in order to add something to his fame and to show his talent in another way, he selected an instrument [the saxophone] completely capable of showing off his ability as a virtuoso . . . Mr Wuille, moreover, produced marvellous effects on the saxophone and possesses the true sound of that instrument, an advantage often denied to clarinettists.³¹

Development

All other instruments have come under notable modifications through time and migration. And last, all have been perfected by slow progress. The saxophone, on the other hand, was born yesterday. It is the fruit of a single conception, and from its first day it has been the same instrument it will be in the future.³²

This was the opinion of François-Joseph Fétis, Belgian musicologist and friend of Sax, writing in his *Biographie universelle des musiciens et bibliographie général de la musique* of 1844. While it may be true that in 1864 the saxophone had indeed changed little since it was first constructed by Sax, there have been a surprisingly large number of modifications since.³³

One of the most obvious changes was the decline nearly to extinction of the orchestral group of saxophones by the end of the nineteenth century. Sax originally conceived the family of saxophones in two groups, each extending from sopranino to contrabass: the original orchestral group in F and C, and the military group alternating in E♭ and B♭. Related to this decline is a second change, as noted by Albert Lavignac: the increasing popularity of the alto saxophone.³⁴ All accounts of the first saxophone indicate that it was of low register; according to Berlioz in 1843, 'the most beautiful low voice

known to this day'. Berlioz more specifically described the earliest saxophone as 'a transposing B $\flat$ instrument, its range is this: contra B $\flat$ to C², including the chromatic tones'.³⁵ Note that Berlioz gives a range of more than three octaves. This alerts us to a third change: by 1855 Berlioz, in his *Treatise upon Modern Instrumentation and Orchestration*, had reduced the range to its more 'normal' two-and-a-half octaves.³⁶

Why did these changes occur? The first, the decline of the orchestral group, probably has many related reasons: the pre-existing establishment of the orchestra as found in the symphonic works of Haydn, Mozart and Beethoven; the German animosity (perhaps led by Wieprecht) towards Sax which precluded the saxophone from the advances in orchestration made by Wagner; and the steadily increasing use of the saxophone in military bands, which favoured the E $\flat$ and B $\flat$ instruments. The second change, the movement away from the bass and baritone saxophones and towards the alto, may have been inevitable: if the future of the instrument depended on public awareness and a growing number of performers, the virtuosity possible on the smaller instrument makes the alto a more logical choice rather than the larger, more expensive and relatively cumbersome bass. The bass saxophone was intended, in part, to fill a void which was not adequately filled by such instruments as the ophicleide and the serpent, but was ably filled by two other new instruments – the tuba and, ironically, Sax's bass clarinet.

The question of a decrease in range is more problematic. We have no reason to doubt that Berlioz (and others) heard a three-octave compass from Sax, but apparently the upper range was infrequently used, and remained so for several decades. Sax assisted Kastner in creating his *Méthode complète et raisonnée de saxophone*, the first saxophone method book, published in 1845. 'Kastner's method book probably contains the truest picture of Sax's conception of saxophone performance practices, since the two men worked out many of the book's details together.'³⁷ The book, 142 pages in length, presents fingerings for a written range of b to f³. Not until the third decade of the twentieth century did the topic reappear in method books and pamphlets by writers such as Bolduc, Eby, Lyon and Winn.

In addition, to further promote the instrument Sax published at least thirty-five compositions by such composers as Arban, Demersseman, Klosé and Singelée. Nineteen of these works have been obtained by Ronkin, who notes that the written range for the saxophone corresponds to the range in the

method book – two-and-a-half octaves from b to f³, with an optional lower alternative often provided for some of the higher passages.

Sax was obviously aware that the saxophone was capable of producing tones above high F. . . Although these high notes are not notated in any of the Sax publications, it is possible that Sax taught altissimo notes to his more advanced students. It would therefore not be unreasonable to assume that these talented students might have employed some high notes in the endings of the Sax publications. However, this is all conjecture. It should be remembered that early reviews of saxophone performance concentrated on the instrument's fluid tone colour and rarely mentioned feats of technical calisthenics or high-note acrobatics.³⁸

Technical changes

In 1860 Sax conceived an extraordinary legal manoeuvre. The impending expiration dates of his patents on the saxotromba and the saxophone, and the enormous costs associated with the lawsuits against them, led Sax to petition for an extension of patent. The petition cited Sax's remarkable innovations and the exceptional circumstances which had prevented him from realising any profit from those patents. For only the second time in French history since 1791, such a request was granted. Patents for the saxotromba and the saxophone each received a five-year extension, giving Sax exclusive rights to the saxophone until 11 May 1866.³⁹ It is a sure indication of the instrument's popularity that other companies presented their own saxophones immediately after the patent came into the public domain.

It was not until 1887, however, that the Evette and Schaeffer Society filed for a patent that included several modifications of lasting significance. Among these modifications were the addition of a connecting bar to permit a trill from F or F[♯] to G[♯], a right-hand chromatic side key for F[♯], the *bis* key to play B[♭] using the first finger of the left hand alone, and the extension of the saxophone range to low B[♭].⁴⁰

A patent granted to A. Lecomte and Company in 1888 provided for a single octave key and rollers for the low E[♭] and C keys. At about the same time the Fontaine-Besson Company made a saxophone with a right-hand side C trill key.⁴¹

Legacy

One is constantly fascinated by the impression which the saxophone made on those of Sax's contemporaries who heard the instrument for the first time.

The finest musicians in Paris were unanimous in their praise of the sheer beauty of the saxophone's sound. Sax gave a demonstration of the saxophone at the Paris Conservatoire in early June of 1842, soon after his arrival in the French capital. Escudier wrote in *La France musicale* of the 'remarkable . . . intensity and quality of sound. . . . You cannot imagine the beauty of sound and the quality of the notes.'⁴² In a *Journal des débats* article which Kochnitzky called the birth certificate of the saxophone,⁴³ Berlioz proclaimed the sound to be incomparable: 'It is full, mellow, vibrant, extremely powerful and [yet] capable of being soft'.⁴⁴

Kastner, in his *Traité général d'instrumentation* of 1844, spoke of 'the nobility and beauty of its timbre. . . . I cannot say enough times, the saxophone is called to the highest destiny by the beauty of its timbre and that opinion is common with several notable musicians, among others Meyerbeer and Halévy, who have heard it the same time as myself'.⁴⁵ In *L'Illustration* of February 1848, we read that 'On hearing it for the first time, Rossini wrote: "This is the most beautiful kind of sound that I have ever heard!" And Meyerbeer said: "It is the beautiful ideal of sound"'.⁴⁶ Liszt declared that 'the ensemble has a really magnificent effect',⁴⁷ an effect which he did not employ but which was used by Charpentier, Halévy, d'Indy, Massenet and Thomas in their orchestral scores before 1894.

Although at the time of Sax's death in 1894 the saxophone had still secured only a tenuous position in the musical world, in just fifty years it had made remarkable progress towards its eventual acceptance. It took the clarinet perhaps a full century from the improvements by Denner to find its way into the Clarinet Concerto and the symphonies of Mozart.⁴⁸ The valved horn was resisted in many quarters and was accepted by the Paris Conservatoire only in 1900, some eighty years after it was first patented by Blümel and Stölzel in 1818.⁴⁹



Fig. 1.4 Quartet of Adolphe Sax saxophones: soprano, *c.* 1859; alto, *c.* 1857; tenor, *c.* 1861; baritone, *c.* 1858

As we have seen, several soloists had achieved recognition on numerous concert stages. In addition, the saxophone had been utilised by many symphonic composers, the vast majority of them French, by 1894. The first such use was by Georges Kastner in *Le Dernier Roi de Juda*. The opera, which included the bass saxophone, received a single performance on 1 December 1844.

Only a pair of works from this fifty-year period have entered the mainstream repertoire: Bizet's well-known *L'Arlésienne Suite No. 1* of 1872 and his *Suite No. 2*, arranged by Ernest Guiraud in 1879. Three operas – *Hamlet* by Ambroise Thomas (1868), and *Hérodiade* (1881) and *Werther* (1892), both by Jules Massenet – should also be mentioned, along with the ballet *Sylvia* by Léo Delibes (1876). Each of these compositions uses the alto saxophone; for *Hérodiade*, Massenet added the tenor and contrabass saxophones. The foundation had been prepared for orchestral works now numbering in the thousands, including those by composers as varied as Béla Bartók, Paul Hindemith, Charles Ives, Zoltán Kodály, Maurice Ravel, Sergei Rachmaninoff, Dmitri Shostakovich, William Walton and Anton von Webern.⁵⁰

To those who associate the saxophone primarily with jazz it is often a surprise to learn that the instrument is a late addition. It is impossible to say when jazz began, but it is clear that the saxophone had gained only an insignificant role at least two decades after jazz began to crystallise.⁵¹ In those first decades of jazz, the typical ensemble consisted of five or six players – trumpet, clarinet, trombone, and a rhythm section of piano, bass and drums.⁵² Very few photographs of jazz musicians before about 1915 include a saxophone, and there is little other evidence to suggest that this eminently well-suited instrument began to exert its extraordinary influence until after World War I.

It was to the band that the saxophone was most enthusiastically welcomed in the nineteenth century. Hemke tells us that the saxophone was accepted in Spanish and English bands in the 1850s, but it was not until 1872 that audiences in the United States were made aware of the effectiveness of the saxophone.⁵³

Patrick Gilmore organised an International Peace Jubilee in Boston and invited the finest bands in Europe to perform between 17 June and 6 July 1872. Among those who accepted the invitation was the French Garde Républicaine Band, who had been sent in a bid to regain some of the prestige

lost by France in the disastrous Franco-Prussian War. The band's fifty-three members included six saxophonists. After the Jubilee, the ensemble performed thirty-two concerts throughout the United States, receiving enthusiastic reviews.⁵⁴ The reviews, in judgements which hold true today, frequently commented on the similarity of the band's sound to that of the orchestra. The saxophones were often cited as one of the principal reasons for this extraordinary sonority.⁵⁵

Shortly thereafter, Gilmore organised the Twenty-Second Regiment Band of New York City. The band's first concert on 18 November 1873 presented saxophone soloist Edouard A. Lefèvre, who remained with the band for nineteen years until Gilmore's death in 1892. Lefèvre headed a section of three saxophones when the band toured Europe in 1878, and a quartet of saxophones for a series of winter concerts in New York that same year. By 1892, Gilmore's band of a hundred musicians included eight saxophonists on soprano, alto, tenor, baritone and contrabass.⁵⁶ After Gilmore's death, Lefèvre joined John Philip Sousa's Band to lead a section of three saxophones (alto, tenor and baritone). Sousa had used such a trio during his years conducting the United States Marine Band between 1880 and 1892.⁵⁷

A later member of the Garde Républicaine Band, already a renowned soloist and creator of the *Quatuor de Saxophones de Paris*, would establish a system of teaching the concert saxophone. Marcel Mule was named Professor of Saxophone at the Paris Conservatoire in 1942, seventy-two years after the previous incumbent – Adolphe Sax. Sax's legacy – the sound of Sax – would make remarkable progress in the twentieth century in the hands of such diverse and gifted musicians as Marcel Mule, Sigurd Rascher, Charlie Parker and John Coltrane.